Expanded response: latent-class faithfulness versus MVN approximation

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Expanded §4.2 response: latent-class faithfulness versus MVN approximation

This note expands the Pass 2 edit applied to §4.2 of `02-dgp-mean-moderation-vs-mvn.tex`, which addressed the reviewer's question in the email summary: if Architecture B is really an 'imperfect responder indicator', why not model that explicitly at the participant level rather than via covariance structure?

The short paragraph added in Pass 2 acknowledged the tension. This expanded treatment develops ten threads, the first five concerning statistical substance and the last five concerning the psychometric and econometric literatures from which the relevant machinery is drawn. The exposition is deliberately paced for readers whose primary training is in biostatistics and who may have had limited contact with latent-variable methods; where psychometric terminology is unavoidable, it is glossed with a biostatistical analogue on first use.

0. Preamble for biostatistician readers

The reviewer's question lives at the interface between two traditions. One is the regression-and-mixed-effects tradition familiar from clinical trial biostatistics, in which participant heterogeneity is represented by random effects and individual response to treatment is estimated from repeated measurements within person. The other is the latent-variable tradition developed within psychometrics and econometrics, in which heterogeneity that is structural (i.e., believed to reflect distinct underlying subpopulations or unobserved traits) is represented by explicit latent classes or latent factors that are estimated jointly with the outcome model. The two traditions study overlapping problems but adopt different conceptual defaults and different notation.

For a biostatistician reading this note, the key translations are the following. A 'latent class' is an unobserved categorical subgroup membership: the closest familiar analogue is the responder/non-responder dichotomy that in ordinary trial practice is only available post hoc through responder analysis with an arbitrary cutoff. A 'latent factor' is an unobserved continuous trait that governs covariation among observed variables: the closest familiar analogue is a frailty in a survival model, or a random intercept that explains correlation across repeated measurements on the same participant. A 'mixture model' is a density that is a weighted sum of simpler densities, where the weights are the unknown class-membership probabilities: the closest familiar analogue is a two-component gaussian where one component represents treatment-responsive participants and the other represents non-responsive participants, both sharing the observed outcome scale. The EM algorithm, which is the standard estimator for these models, is the same algorithm that biostatisticians encounter in multiple imputation and in missing-data maximum likelihood for mixed models; the only conceptual difference is that the 'missing data' being imputed in a mixture fit is the unobserved class label Z_i for each participant rather than a missing covariate or outcome value.

Against this background, the question Architecture B answers is narrower than it may appear. Architecture B does not ask whether discrete responder classes exist; it asks whether the second-moment consequences of unobserved responder classes, in particular the biomarker-by-response covariance that a mixture model would induce, can be captured by a single multivariate normal with treatment-state-dependent correlation. Sections 1-5 below argue that the answer is generally yes for power calculation purposes, with specific caveats. Sections 6-10 place this answer within the broader psychometric spectrum and clarify what a more faithful generative model would look like, drawing on latent-class, factor mixture, and heterogeneous-random-effects literatures that have already worked through the identifiability and estimation problems in detail.

0.1 Notation and term crosswalk

The following shorthand is used throughout. B_i denotes the biomarker value for participant i; in the prazosin-PTSD application, this is a baseline blood-pressure summary. BR_{it} denotes the biological-response component of outcome for participant i at time t; this is one of three latent response factors in the Hendrickson decomposition alongside a time-variant component TV_{it} and a placebo-belief component PB_{it}. D_{it} denotes drug state at time t, with the continuous-decay analysis form $D_{it} = \exp(-\lambda t_{sd,it})$ during off-drug periods and $D_{it} = 1$ during on-drug periods. Z_i denotes an unobserved class label.

The principal psychometric-to-biostatistical term correspondences, given as a running glossary rather than a table to accommodate the longer biostatistical paraphrases:

- **Latent class.** An unobserved categorical subgroup (biostatistical analogue: a true responder versus non-responder stratum that is not directly observed).
- **Latent factor.** An unobserved continuous trait that governs covariation among observed variables (analogue: a frailty in a survival model, or a random effect in a linear mixed model).
- **Class-conditional distribution f_z.** The outcome distribution within a single subgroup (analogue: a subgroup-specific sampling distribution).
- **Mixing proportion π.** The prevalence of a given subgroup in the population (analogue: a stratum proportion).
- **Class indicator function.** Membership in an unobserved stratum, treated as a random variable.
- **Measurement invariance.** Whether a model applies identically across subpopulations (analogue: the assumption underlying pooled versus stratum-specific analyses).
- **Growth mixture model (GMM).** A longitudinal mixed model in which the population of random effects is itself a finite mixture (analogue: a longitudinal LME with subgroup-specific trajectory classes).
- **Factor mixture model (FMM).** A hybrid model with discrete subgroups at the upper level and within-subgroup continuous heterogeneity at the lower level (analogue: a two-level hierarchical model where the upper level is categorical and the lower level is a standard LME).
- **Mixture of experts, regression mixture.** Covariate-dependent class probabilities combined with class-specific regression slopes (analogue: a propensity-score-like gating model paired with within-stratum outcome regressions).
- **EM algorithm.** The same expectation-maximisation familiar from multiple imputation, applied here to the missing class labels Z_i rather than to missing covariates or outcomes.

With this vocabulary in hand, the technical development follows.

1. What the faithful generative model actually looks like

If blood pressure (or any candidate predictive biomarker) genuinely indexes an unobserved responder/non-responder dichotomy, the mechanistically correct data-generating process (DGP) is a finite mixture:

$$Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{it}),$$

where Z_i is the latent class indicator (responder vs. non-responder), $\pi(B_i) = \Pr(Z_i = 1 \mid B_i)$ is typically modelled as a logistic function of the biomarker, and the class-specific response distributions f_0 and f_1 differ in their drug-response parameters (peak response, onset rate, or both).

Unpacking this for readers unfamiliar with mixture notation, the generative recipe is the following. For each participant i, first draw a latent label Z_i that equals 1 with probability $\pi(B_i)$ and 0 otherwise. Participants with higher biomarker values have a higher chance of being drawn into the responder class but not a deterministic assignment, which is why the biomarker is described in the main text as an 'imperfect indicator' of the underlying responder status. Once Z_i is drawn, the within-person trajectory BR_{it} is generated from the response curve appropriate to that class: responders receive a curve with a substantial drug effect; non-responders receive a curve that is essentially flat in D_{it}. Carryover, modelled in this framework through the continuous drug indicator D_{it}, operates only on the class-specific mean trajectories. It does not erode the biomarker-outcome association at the level of the marginal distribution, because the biomarker-mediated mechanism is class membership Z_i, which is fixed over time and drug-state-invariant.

This feature (class membership being carryover-invariant) is the intuitive kernel of the reviewer's concern. Under a faithful mixture DGP, carryover washes out the within-class drug effect but leaves the between-class mean separation intact. Any analysis model that can see class membership, even imperfectly, retains a signal proportional to the between-class difference. By contrast, Architecture B encodes the biomarker-response association purely as a second-moment object (a covariance), and carryover erodes that covariance directly. The two DGPs therefore predict different patterns of power loss as a function of carryover intensity, and this is what Section 3 of the main document documents.

2. Why MVN differential correlation is nonetheless a defensible second-moment summary

A two-component mixture with class-dependent means and a common within-class covariance produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments match, to leading order, those of a single multivariate normal with $\text{Cor}(B, BR) = c_{bm}$ chosen so that c_{bm}^2 equals the between-class variance fraction (the proportion of total variance in BR that is attributable to class separation rather than to within-class residual variation).

The biostatistical intuition is the following. If responders and non-responders differ by some amount Δ in mean drug response and the biomarker has some accuracy $\pi(B_i)$ in predicting class membership, then marginally (i.e., averaging over the unobserved Z_i) there is a linear association between B_i and BR_{it} whose strength is determined by two factors: how far apart the two classes are in drug response (Δ) and how well the biomarker discriminates them (π). A single multivariate normal with an appropriately chosen correlation coefficient c_{bm} can reproduce this marginal linear association even though no such gaussian actually generated the data. This is the sense in which Architecture B 'approximates' the mixture DGP: it reproduces the marginal biomarker-response association that a mixture would produce, without committing to the discrete generative mechanism.

Under mild regularity conditions (adequate class overlap and moderate class-prevalence probabilities π away from 0 and 1), the mixture's second-moment structure is well-approximated by an MVN with differential correlation: high correlation in treatment states where class-specific means separate (i.e., on-drug) and shrinkage towards zero in treatment states where class-specific means coincide (i.e., off-drug after full carryover washout). Architecture B as implemented therefore captures the covariance implications of the latent-class mechanism without committing to the discrete generative form. For power calculations driven by linear association statistics (t-tests for $\beta_{bm:D}$ in a linear mixed model), this approximation is often adequate.

3. Where the approximation breaks down

The MVN approximation will be misleading in three identifiable regimes, each of which has a concrete biostatistical signature.

- **Bimodal responses.** When f_0 and f_1 are well-separated (non-responders show near-zero drug effect and responders show a large effect), the marginal distribution of BR is bimodal: a histogram of on-drug response would show two peaks rather than a single peak with gaussian scatter around it. No MVN matches the tails of a bimodal distribution, and power under the true mixture DGP will be higher than Architecture B predicts because a well-specified mixture analysis can exploit the bimodality. Biostatisticians working with clinical response data will recognise this pattern from settings in which a subpopulation genuinely does not respond to a drug (e.g., certain targeted oncology treatments), where response histograms are characteristically non-gaussian.
- **Strong class-membership gradient.** When $\pi(B_i)$ is steep (approaching a step function at some biomarker threshold), the biomarker behaves almost deterministically as a class label. Architecture A (mean moderation) with a thresholded or dose-response moderation then becomes the better approximation, because the biomarker-response relationship is dominated by its mean-structure component. The familiar biostatistical scenario here is a diagnostic biomarker with near-perfect sensitivity and specificity for the responsive phenotype, where dichotomising the biomarker at its clinical threshold would lose little information.
- **Class-varying covariance.** If the autocorrelation, residual variance, or placebo-response parameters themselves differ between classes (i.e., responders and non-responders differ not only in mean drug effect but in how variable their day-to-day symptom scores are, or in how autocorrelated those scores are), no single-component MVN can represent the generative structure. Analysis models assuming constant covariance will be misspecified regardless of whether they model the biomarker interaction in the mean or in the covariance. This is the latent-class analogue of the heteroscedastic residual problem that biostatisticians address with `varIdent` or `weights` arguments in `nlme::lme`, except that the stratifying variable is itself unobserved.

4. Implications for the power estimates in Section 3

The substantial Architecture B power loss under carryover (40-60% across the designs reported in Section 3 of the main document) reflects erosion of a second-moment signal. A mixture-model analysis fitted to mixture-DGP data would, in principle, recover additional power by estimating class membership directly, because the class label is carryover-invariant. The gap between Architecture B's carryover-sensitive power estimate and the mixture-faithful power estimate is an upper bound on what a mixture analysis could recover at the same carryover level; whether this gap is large in the prazosin-PTSD parameter regime is an empirical question not yet answered by the existing pmsimstats infrastructure.

A focused simulation contrasting three cells would characterise the loss attributable to second-moment approximation versus the loss attributable to analysis-model misspecification:

1. Mixture DGP analysed with the current MVN linear mixed model (i.e., the analysis model makes no attempt to recover class structure).
2. Mixture DGP analysed with a matched mixture analysis model (e.g., a `lcmm`-style joint latent class mixed model).
3. MVN DGP (Architecture B as currently implemented) analysed with both the current MVN linear mixed model and the `lcmm`-style mixture model, to verify that the mixture model does not gain power on data for which no

latent class structure exists.

The difference between cells 1 and 2 quantifies the power recovery attributable to using a faithful analysis model; the difference between cells 1 and 3 (with MVN analysis) quantifies the approximation error in Architecture B itself. This factorial design maps cleanly onto the existing pmsimstats simulation harness and would be a natural extension of the present work.

5. Practical recommendation and scope of the open question

For the prazosin-PTSD application, mixture modelling is attractive in principle but carries identifiability costs that are acute in N-of-1 settings with modest participant counts. The EM algorithm for finite mixtures of mixed-effects models requires either informative priors on class-membership probabilities or strong separation between class-specific response curves, neither of which is assured in a trial designed to detect an interaction whose existence is itself uncertain. The standard diagnostic for mixture identifiability, the entropy of posterior class-membership probabilities, typically requires sample sizes in the hundreds before it stabilises at values that support confident class extraction (Bauer and Curran 2003, Bauer 2007). Aggregated N-of-1 trials of the size contemplated in the prazosin-PTSD application (typically $N \in [30, 100]$) are an order of magnitude below this threshold.

The open question, therefore, is not whether mixture modelling is more faithful (it is), but whether the fidelity gain is recoverable at realistic sample sizes, or whether analysis-strategy mitigations (Section 6 of the main document: restricted analysis, weighting, within-subject contrasts) deliver a better power-per-complexity trade. These analysis-strategy mitigations are closer to standard biostatistical practice and have better-understood inferential properties at the relevant sample sizes. Resolving the mixture-versus-mitigation question empirically is out of scope for the present white paper but is a natural next step in the pmsimstats programme, and the factorial simulation outlined in Section 4 above would provide the evidence base required.

6. Connection to the psychometric latent variable literature

The reviewer's question, translated into psychometric vocabulary, asks whether biomarker-moderated treatment response is better modelled as a continuous-trait phenomenon (classical factor-analytic or latent regression tradition) or as a discrete-class phenomenon (latent class or mixture tradition). The psychometric literature has worked this distinction over for more than fifty years, originally in the context of individual differences in education, personality, and psychopathology, and the taxonomy it developed maps directly onto the Architecture A versus Architecture B distinction drawn in this paper.

The core generative question is whether unobserved heterogeneity in outcome distributions is best represented by (i) a single population with continuous individual-differences variation along one or more latent dimensions (factor-analytic models; Spearman 1904; Lawley and Maxwell 1971), (ii) a small number of qualitatively distinct subpopulations with within-class homogeneity (latent class models; Lazarsfeld and Henry 1968; Goodman 1974), or (iii) a hybrid in which classes differ along continuous dimensions within each class (factor mixture models; Lubke and Muthén 2005; Clark et al. 2013). For biostatistician readers, the three alternatives correspond respectively to: a standard linear mixed model with a continuous random effect (option i), a stratified analysis with unobserved strata (option ii), and a stratified analysis with within-stratum random effects (option iii). The same trichotomy appears throughout the clinical trial simulation literature, usually implicitly: Architecture A encodes the continuous individual-differences view (treatment response is a deterministic function of a continuous biomarker), while Architecture B as implemented in Hendrickson et al. (2020) is most naturally interpreted as a second-moment approximation to the latent-class view.

Recognising this connection has practical consequences for the pmsimstats programme. The decision between Architecture A and Architecture B is not merely a parameterisation convenience; it reflects a substantive biological hypothesis about whether biomarker-outcome coupling arises from graded continuous variation (Architecture A, consistent with a mechanism in which every participant has some degree of drug response, scaled by biomarker) or from discrete subpopulation structure (Architecture B in its latent-class interpretation, consistent with a mechanism in which some participants respond substantially and others not at all, with the biomarker predicting which are which). The two hypotheses yield comparable marginal biomarker-outcome associations at a single timepoint but diverge sharply under carryover, which is the core finding of Section 3.

7. Finite mixture taxonomy relevant to biomarker moderation

Four families of finite mixture model from the psychometric and econometric literatures bear directly on the biomarker-treatment interaction problem. The exposition below introduces each family with its biostatistical analogue, its psychometric origin, and its specific relevance to the pmsimstats framework.

7.1 Latent class and latent profile analysis

Latent class analysis (LCA) and the closely related latent profile analysis (LPA) are mixture models for cross-sectional data in which each participant belongs to one of a small number of unobserved classes, and the observed variables have class-specific marginal distributions. LCA is the version with categorical indicators (e.g., symptom-present / symptom-absent items in a psychiatric assessment), and LPA is the version with continuous indicators (e.g., biomarker measurements or symptom severity scores).

The biostatistical reader should picture this as follows. Suppose one wished to identify a responder subgroup using baseline characteristics alone, without access to any treatment-response information. If the responder subgroup has a characteristic baseline biomarker profile that differs from the non-responder profile, a mixture model fitted to the baseline biomarker set can estimate (a) the proportion of participants in each class, (b) the class-specific mean biomarker profiles, and (c) each participant's posterior probability of belonging to each class. These posterior probabilities function as soft subgroup-membership predictors that can subsequently be used as covariates in outcome analysis. This is the two-stage analytic strategy standard in psychometric applications. In the biomarker-trial context, an LPA-style model would posit two or more responder classes with class-specific marginal distributions of on-drug and off-drug response, and estimate class-membership probabilities from the observed response trajectories rather than from baseline covariates alone.

Classical references include Lazarsfeld and Henry (1968), Goodman (1974), Clogg (1995), and McLachlan and Peel (2000); the Vermunt and Magidson Latent GOLD software and the `poLCA` R package implement the standard EM estimation routines. For a biostatistician coming to this literature, McLachlan and Peel (2000) is the most accessible entry point, as it is structured as a statistical monograph rather than a substantive-psychometric one.

7.2 Growth mixture models and group-based trajectory models

Growth mixture models (GMM; Muthén and Shedden 1999; Muthén 2001, 2004) extend LPA to longitudinal data by specifying class-specific growth (or, in the trial setting, drug-response) curves with within-class random-effects variation. In biostatistical vocabulary, a GMM is a longitudinal linear mixed model in which the population distribution of random effects is itself a mixture: rather than assuming a single multivariate normal for random intercepts and slopes, GMM assumes a mixture of multivariate normals, each corresponding to a distinct trajectory class. Group-based trajectory models (Nagin 1999, 2005) are the degenerate case in which within-class random-effect variances are set to zero, so that all participants within a class share an identical trajectory up to observation-level residual noise.

GMM is the closest psychometric analogue of the mixture DGP discussed in §4.2: each participant belongs to a latent class with its own drug-response trajectory, and the class label is drug-state-invariant. For the pmsimstats biomarker moderation setting, a GMM that uses the biomarker as a class-membership predictor is essentially the faithful generative model described in Section 1 of this note. Bauer and Curran (2003) and Bauer (2007) document the identifiability and overextraction hazards that accompany GMM estimation at realistic sample sizes, showing in particular that GMM can extract apparent class structure from data generated by single-class non-gaussian processes. These cautions apply with full force to N-of-1 trials, where the number of participants is typically well below the sample sizes at which GMM class-count recovery has been validated.

7.3 Factor mixture models

Factor mixture models (FMM; Yung 1997; Arminger, Stein and Wittenberg 1999; Lubke and Muthén 2005) combine continuous latent factors with discrete latent classes, yielding a generative structure in which within-class variation is governed by a factor model and between-class variation is governed by class-specific means and (optionally) class-specific factor loadings or residual covariances. The FMM framework subsumes both the pure continuous (single-class factor model) and the pure discrete (LCA with no within-class structure) cases as limits.

For biostatisticians, the practical intuition is that FMM is a two-level hierarchy: at the upper level, participants are assigned to one of a small number of classes (as in LCA); at the lower level, each class has its own within-class random-effects structure (as in a standard linear mixed model). This corresponds to the biological hypothesis that there are genuinely distinct responder subtypes, and within each subtype there is additional continuous heterogeneity driven by covariates or unmeasured individual differences. For the biomarker-moderation problem, FMM provides a principled way to represent a biomarker as both a noisy class indicator (via class-membership probability) and a continuous modulator of within-class response (via factor loadings on the biological-response factor). This hybrid representation is a better fit to the biology that motivates the prazosin-PTSD application than either pure LCA or pure factor analysis, but estimation requirements are correspondingly higher.

7.4 Regression mixtures and mixtures of experts

Regression mixtures (DeSarbo and Cron 1988; Wedel and DeSarbo 1995; Jedidi, Jagpal and DeSarbo 1997) and the closely related mixture of experts architecture (Jacobs, Jordan, Nowlan and Hinton 1991) allow class-specific regression coefficients with class-membership probabilities that depend on covariates (the 'gating' function in mixture-of-experts terminology). For a biostatistician, the clearest way to see this family is through its relationship to standard interaction modelling. An ordinary linear model with a treatment-by-biomarker interaction posits a single regression surface in which the slope of response on treatment varies smoothly with the biomarker. A regression mixture instead posits two or more distinct regression surfaces (one per class), with a separate covariate-dependent model governing which surface applies to which participant. The two-stage structure (classification by the gating model, then prediction by the within-class regression) is directly analogous to the two-stage structure of causal inference methods that use propensity scores to stratify before estimating within-stratum treatment effects.

In the trial-simulation idiom, regression mixtures correspond to a DGP in which the biomarker governs both the probability of being a responder and the magnitude of the within-responder drug effect, which is precisely the generative form the reviewer's question invites. `flexmix` in R (Leisch 2004; Grün and Leisch 2008) implements this family with user-definable component models, including mixed-effects components suitable for repeated-measures data.

8. The Architecture B spectrum: covariance, mean, and combined biomarker moderation

Architecture B as implemented in Hendrickson et al. (2020) and pmsimstats specifies biomarker-dependent covariance structure with population means held constant across biomarker values. This is one point on a broader spectrum of biomarker-moderated DGPs recognised in the psychometric literature; the spectrum is usefully organised by which moments of the joint distribution depend on the biomarker. For biostatistician readers, the moment-by-moment organisation is the natural way in: any multivariate normal distribution is characterised by its mean vector and covariance matrix, and biomarker moderation can in principle enter either object or both.

8.1 Covariance-only moderation (Architecture B as implemented)

The Hendrickson-Schork specification holds $E[BR_{it}]$ fixed across biomarker strata (i.e., high- and low-biomarker participants have the same expected drug response at any given time) and encodes the interaction entirely in $\text{Cov}(B_i, BR_{it} \mid D_{it})$ (i.e., high- and low-biomarker participants differ in the covariance between biomarker and drug response, specifically in how strongly the biomarker co-varies with response during on-drug periods versus off-drug periods).

This is an unusual restriction within the psychometric mixture tradition. Most standard FMM and GMM parameterisations permit class-specific means at a minimum, and the means-constant-covariance-varies case is typically encountered only as a constrained submodel used for testing invariance hypotheses (Meredith 1993; Widaman and Reise 1997). The biostatistical analogue is a model in which two subgroups have identical expected outcomes but differ in residual variance or in within-subject autocorrelation: such models are occasionally fitted in sensitivity analyses but rarely assumed as the primary DGP. Architecture B's appeal in the Hendrickson context is its tractability for power simulation under multivariate normality (the entire DGP can be reduced to drawing from a single MVN with a cleverly constructed covariance). Its cost, as the reviewer observes, is that it does not correspond cleanly to any natural generative mechanism.

8.2 Combined mean and covariance moderation

Arminger, Stein and Wittenberg (1999), in the canonical Psychometrika treatment of mixtures with covariate-dependent structure, develop finite mixtures of conditional mean- and covariance-structure models in which both the expected-response vector and the residual covariance matrix may differ between classes and may depend on observed covariates within class. This is the general psychometric form of Architecture B and reduces to Architecture A in the limit of a single class with covariate-dependent mean only, and to the Hendrickson-implemented submodel in the limit of class-invariant means with covariate-dependent covariance only.

The biostatistical interpretation of the combined form is that responders and non-responders differ both in their typical drug response (mean moderation, Architecture A) and in how noisily that response is expressed within class (covariance moderation, Architecture B). Both channels carry information about class membership: the mean-moderation channel contributes via the conditional expectation $E[BR \mid B]$ and the covariance-moderation channel contributes via the conditional variance $\text{Var}[BR \mid B]$. Under carryover, the mean-moderation channel is comparatively robust because class means need not converge off-drug; the covariance-moderation channel is eroded because off-drug covariance is by construction attenuated. A combined-specification simulation would therefore be expected to show power loss intermediate between the Architecture A and Architecture B extremes, and this intermediate case is arguably the most biologically plausible regime for the prazosin-PTSD application. Verbeke and Lesaffre (1996) provide the analogous treatment for linear mixed-effects models with heterogeneity in the random-effects population, which is the more direct generalisation of the repeated-measures structure used in pmsimstats and is the entry point most likely to feel familiar to biostatistician readers.

8.3 Location-scale moderation (GAMLSS and related)

Generalised additive models for location, scale, and shape (GAMLSS; Rigby and Stasinopoulos 2005) and the related distributional regression tradition (Klein et al. 2015) allow the biomarker to enter multiple parameters of the outcome distribution simultaneously through separate link functions. In plain terms, a standard regression models the conditional mean of an outcome as a function of covariates; a GAMLSS-style distributional regression models the conditional mean, the conditional variance, and if appropriate the conditional skewness and kurtosis, each as a separate function of covariates. For biostatisticians, the familiar entry point is the heteroscedastic linear regression in which the residual variance is allowed to depend on a covariate; GAMLSS generalises this to arbitrary distributional shape.

In the biomarker-trial context, a GAMLSS-based DGP would allow high-biomarker participants to have both a shifted mean drug response (a location effect, which is Architecture A) and altered response variability (a scale effect, which is analogous to Architecture B's covariance moderation but operating on marginal rather than joint moments), without requiring discrete class structure. The GAMLSS framework is thus the continuous-heterogeneity analogue of FMM and occupies an intermediate position between pure Architecture A (mean only) and pure Architecture B (covariance only). This intermediate position is attractive for sensitivity analysis because it does not require committing to a specific number of latent classes.

8.4 Heterogeneous random-effects models

Verbeke and Lesaffre (1996), Zhang and Davidian (2001), and Proust-Lima et al. (2017) develop linear mixed-effects models in which the random-effects distribution is itself a finite mixture, allowing class-specific random-intercept and random-slope distributions. The biostatistical setting this extends is the standard linear mixed model with random intercept and random treatment slope, familiar from any longitudinal trial analysis using `nlme::lme` or `lme4::lmer`. In the standard model, the random treatment slope is drawn from a single normal distribution with mean equal to the average treatment effect and variance capturing individual heterogeneity. In the heterogeneous random-effects extension, that single distribution is replaced by a mixture: a fraction of participants are drawn from a 'responder' distribution with a large mean treatment slope and another fraction from a 'non-responder' distribution with a near-zero mean treatment slope. The biomarker predicts which mixture component a participant comes from.

Applied to aggregated N-of-1 trial data, this framework represents responder heterogeneity directly at the participant-specific random treatment effect, which is arguably the most clinically transparent representation available: responders and non-responders have distinct random-slope distributions on the treatment indicator, and the biomarker predicts class membership rather than entering the analysis-model mean structure at the fixed-effect level. The `lcmm` R package (Proust-Lima et al. 2017) implements this family and is the closest available tooling to the pmsimstats use case: it accepts standard longitudinal mixed-model syntax, fits a mixture on the random-effects distribution, and returns posterior class probabilities for each participant. For the biostatistician planning an extension of the current pmsimstats simulation, `lcmm` is the recommended entry point.

9. Implications for the pmsimstats framework

Three implications follow from locating Architecture B within the broader psychometric spectrum; each is stated first in summary form and then unpacked in biostatistical terms.

First, the covariance-only parameterisation adopted in Hendrickson et al. (2020) is the most restrictive operational case of a much richer family. The substantial power loss under carryover documented in Section 3 is a specific property of this restrictive case and may not generalise to the mean-plus-covariance or heterogeneous-random-effects variants that are standard in psychometrics. Unpacked, this means that a sensitivity analysis based solely on Architecture B risks overstating the carryover-induced power loss that a biomarker-guided aggregated N-of-1 trial should expect, because the DGP on which Architecture B is built puts all of the biomarker-outcome signal into a channel (the covariance) that carryover erodes. If, in reality, some of the biomarker-outcome signal flows through the mean structure (as in Architecture A or in the combined specification of §8.2), that component of the signal survives carryover, and overall power loss is smaller than Architecture B predicts. A complete sensitivity analysis for the prazosin-PTSD application should accordingly report power under at least three points on the spectrum: Architecture A (mean-only), Architecture B as implemented (covariance-only), and a combined mean-plus-covariance specification of the Arminger et al. (1999) form. The existing pmsimstats infrastructure already supports the first two; adding the third requires a modest extension to `generateData` that permits simultaneous mean scaling and covariance moderation.

Second, the identifiability concerns documented in the psychometric GMM and FMM literatures (Bauer and Curran 2003; Bauer 2007; Lubke and Muthén 2005) bound what can be asked of mixture-based analysis models at N-of-1 trial sample sizes. The published FMM applications typically involve several hundred to several thousand participants; biomarker-trial applications with $N \in [30, 100]$ are well outside this range. For a biostatistician, this means that analysis plans built around fitting an `lcmm` or `flexmix` mixture model directly to trial data are unlikely to yield reliable class-membership inference and should not be relied upon as the primary analysis. Analysis-strategy mitigation (Section 6 of the main document) is therefore more likely than mixture-model analysis to deliver recoverable power gains at realistic trial sizes. Mixture modelling remains useful as a sensitivity check (to see whether class-structured residuals are consistent with the pre-specified analysis) and as a DGP for simulation (to verify robustness of the primary analysis to class-structured departures from gaussianity).

Third, if the pmsimstats programme pursues a richer DGP in subsequent work, the natural target is the heterogeneous-random- slopes form (§8.4) rather than a full FMM. The random-slopes parameterisation preserves the analysis model's compatibility with existing N-of-1 estimation machinery (`nlme::lme` with participant-specific random treatment effects) while admitting the biologically faithful latent-class generative structure at the DGP level. For a biostatistician, this is the most conservative path forward: the simulation DGP is extended in a familiar direction (a mixture over random-slope distributions, which is a natural relaxation of the homogeneous-population assumption built into standard mixed-effects models) while the analysis model remains the pre-specified `nlme::lme` fit. This preserves the audit chain of the current framework and permits direct comparison with the covariance-only baseline under carryover.

10. Software and implementation references

R packages relevant to estimating the models surveyed above, in approximate order of fit to the pmsimstats use case:

- `lcmm` (Proust-Lima et al. 2017). Joint latent class mixed models for longitudinal and survival data; the closest available tooling for heterogeneous random-effects specification (§8.4). For the biostatistician coming to this literature, `lcmm`'s syntax is essentially `nlme::lme` with a latent-class extension, making it the lowest-barrier entry point.
- `flexmix` (Leisch 2004; Grün and Leisch 2008). General regression mixture framework with user-definable component models; suitable for regression-mixture and mixture-of-experts specifications (§7.4). `flexmix` is more general than `lcmm` but requires more configuration.
- `mclust` (Scrucca et al. 2016). Gaussian finite mixture models with a range of class-specific covariance parameterisations; supports the Arminger et al. (1999) mean-plus-covariance specifications non-longitudinally. `mclust` is useful for cross-sectional LPA-style analyses of baseline biomarker panels but does not natively

handle repeated-measures structure.

- `gamlss` (Rigby and Stasinopoulos 2005). Distributional regression with covariate-dependent location, scale, and shape parameters (§8.3). For biostatisticians familiar with `mgcv`, the syntax and model-fitting workflow of `gamlss` will be largely familiar.
- `poLCA` (Linzer and Lewis 2011). Polytomous latent class analysis for the categorical-indicator case. Less directly relevant to the pmsimstats continuous-outcome setting, but useful if baseline class indicators (e.g., symptom-checklist items) are incorporated into class assignment.
- `OpenMx` and `lavaan` (with mixture extensions). Structural equation mixture modelling for FMM-style specifications; more general but with steeper configuration cost than `lcmm` or `flexmix`.

The psychometric reference implementation outside R is Mplus (Muthén and Muthén 1998-2017), which remains the standard for GMM, FMM, and related mixture-SEM specifications. Most of the psychometric literature on mixture-model identifiability and overextraction hazards (Bauer and Curran 2003; Lubke and Muthén 2005) uses Mplus for its primary simulations, and investigators cross-validating pmsimstats simulation results against published psychometric benchmarks may need occasional access to Mplus for direct comparison.

Relation to the existing §4.2 paragraph

The five threads in Sections 1-5 above could extend the §4.2 Pass 2 paragraph in place, or be promoted to a new §4.3 ('Mixture-modelling alternative to MVN differential correlation') so that §4.2 retains its current clinical-examples focus. Either approach preserves the document's existing architecture-comparison narrative while giving the reviewer's concern the technical treatment it warrants.

The psychometric-connection material in Sections 6-10 is more naturally placed either as a new §4.4 ('Architecture B in the psychometric latent-variable tradition') or as an appendix. It answers a distinct question from the §4.2 edit: whereas §4.2 concerns the biological faithfulness of Architecture B, Sections 6-10 concern its methodological provenance and the spectrum of related specifications already developed in the psychometric literature. Treating the two as separate document additions preserves the logical structure of Section 4 (biological assumptions) while giving the methodological-provenance discussion the space it requires.

For a biostatistician preparing the manuscript revision, the recommended sequencing is: incorporate Sections 1-5 into the main response to reviewer (as §4.3), defer Sections 6-10 to a methodological appendix that can be cited but need not be read for the primary argument to go through, and commit to the factorial simulation described in Section 4 as a follow-on paper. This preserves the white paper's focus on the prazosin-PTSD application while placing the broader methodological context on record for future reviewers and collaborators.

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